

LAUGHLIN AFB, TX, USA

WMO: 722615

Lat: 29.367N

Lon: 100.783W

Elev: 330

StdP: 97.43

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 22001

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.8	1.1	-12.6	1.3	10.8	-10.2	1.6	10.5	12.6	15.5	11.2	13.1	2.9	10	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.2	40.2	22.2	38.9	22.5	37.8	22.6	26.2	31.8	25.4	31.7	24.9	31.4	5.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	20.8	28.5	23.9	19.6	27.6	23.3	18.8	27.2	83.5	31.7	79.7	31.7	77.4	31.3	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.2	8.3	DB	-4.6	41.6	2.7	1.9	-6.6	42.9	-8.2	44.0	-9.7	45.0	-11.7	46.4	(4)
(5)			WB	-6.2	27.2	2.3	1.7	-7.9	28.3	-9.2	29.3	-10.6	30.2	-12.3	31.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.1	11.9	15.1	18.6	22.7	26.6	29.6	30.7	30.9	27.6	22.6	16.8	12.3	(6)	
	DBStd	7.80	4.70	5.25	4.66	3.90	3.45	2.38	2.43	2.06	3.00	4.20	4.41	4.39	(7)	
	HDD10.0	85	35	15	3	0	0	0	0	0	0	1	5	26	(8)	
	HDD18.3	671	206	113	55	9	1	0	0	0	0	14	79	193	(9)	
	CDD10.0	4514	93	157	271	380	514	589	642	647	527	390	208	97	(10)	
	CDD18.3	2059	4	22	64	139	257	339	383	388	278	146	32	5	(11)	
	CDH23.3	25023	69	234	571	1401	2921	4440	5299	5364	3151	1295	221	58	(12)	
(13)	CDH26.7	13439	11	77	199	613	1497	2513	3165	3202	1616	498	41	8	(13)	
(14)	Wind	WSAvg	4.2	3.6	4.0	4.5	4.9	5.1	4.7	4.3	3.6	3.9	3.6	3.4	(14)	
(15) Precipitation	PrecAvg	523	19	18	35	36	70	56	55	60	75	53	29	17	(15)	
	PrecMax	881	60	71	93	130	219	153	175	262	194	170	133	78	(16)	
	PrecMin	246	1	0	0	0	5	1	1	2	1	3	0	0	(17)	
	PrecStd	168	18	18	27	36	57	45	55	75	51	52	31	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.9	32.5	33.6	37.4	40.4	41.3	42.3	41.1	40.2	35.1	29.9	27.5	(19)	
		MCWB	14.3	15.5	17.0	18.8	20.3	21.8	22.3	22.5	22.1	21.3	18.2	15.4	(20)	
	2%	DB	25.2	28.9	31.2	34.8	37.7	39.1	40.2	39.4	37.4	32.9	27.8	24.9	(21)	
		MCWB	14.2	15.0	17.0	19.1	21.5	22.7	22.3	22.8	22.4	21.5	17.7	16.0	(22)	
	5%	DB	22.8	26.2	28.9	32.4	35.7	37.5	38.8	38.4	35.8	31.3	26.0	22.5	(23)	
		MCWB	13.2	15.8	17.3	19.2	21.7	22.9	22.6	22.9	22.4	20.9	17.6	15.2	(24)	
	10%	DB	20.3	23.6	26.7	30.2	33.7	36.1	37.4	37.4	34.0	29.6	24.0	20.2	(25)	
		MCWB	12.7	14.8	17.1	18.6	21.5	22.8	23.0	22.8	22.1	20.6	17.1	14.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.9	20.7	22.3	24.8	26.8	27.5	27.8	26.2	25.7	25.1	22.9	20.7	(27)	
		MCDB	22.3	24.6	25.9	27.6	29.1	32.4	34.7	33.8	31.9	30.1	25.5	22.9	(28)	
	2%	WB	17.1	19.5	21.0	23.0	25.2	26.1	25.8	25.3	25.0	24.0	21.2	18.9	(29)	
		MCDB	21.3	23.2	25.4	28.7	30.0	31.9	32.8	32.7	31.1	29.0	23.9	21.5	(30)	
	5%	WB	15.7	18.4	20.1	22.1	24.4	25.3	25.0	24.8	24.4	23.3	20.1	17.2	(31)	
		MCDB	19.8	21.9	24.5	27.6	30.1	31.4	32.1	32.2	30.2	27.8	23.0	20.1	(32)	
	10%	WB	14.0	17.1	19.2	21.4	23.7	24.7	24.4	24.3	23.9	22.5	18.9	15.4	(33)	
		MCDB	18.4	21.3	23.7	26.7	29.7	31.1	31.6	31.8	29.6	26.9	22.6	18.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	13.5	13.0	13.4	12.2	12.0	12.0	12.2	11.6	12.1	12.1	12.7	(35)	
		MCDBR	17.4	16.8	16.0	16.0	14.9	13.8	13.9	13.6	13.4	13.4	13.7	16.3	(36)	
	5% WB	MCWBR	9.1	8.4	7.3	7.1	5.1	4.0	3.9	3.8	4.0	5.4	7.1	8.9	(37)	
		MCWBR	13.1	11.7	12.1	11.7	11.5	11.2	11.5	11.8	10.4	9.9	10.0	11.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.299	0.304	0.326	0.373	0.391	0.400	0.411	0.392	0.383	0.344	0.319	0.302	(40)	
		taud	2.557	2.550	2.501	2.384	2.387	2.407	2.413	2.464	2.431	2.538	2.536	2.550	(41)	
	Ebn at Noon		947	970	963	921	899	886	875	891	892	913	915	926	(42)	
		Edh at Noon	85	93	104	122	122	119	118	111	111	94	86	82	(43)	
(44) All-Sky Solar Radiation	RadAvg		3.29	4.04	5.05	5.89	6.10	6.87	6.81	6.51	5.34	4.35	3.46	2.98	(44)	
	RadStd		0.38	0.56	0.46	0.56	0.59	0.52	0.58	0.36	0.46	0.59	0.43	0.39	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	-0.84	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page